

Homework — 2.1.3 Thinking Procedurally

Name: _____ Date: _____ Mark: _____ / 20

OCR H446 — set after the 2.1.3 lesson.

Part A — This week's topic (~14 marks)

1. Write the letter of the correct definition in the Match column for each term. Two definitions are not needed. [2] (AO1)

Term	Match
Decomposition	
Sub-procedure	
Dependency	
Independent steps	

Letter	Definition
A	Breaking a complex problem down into smaller, more manageable sub-problems
B	Steps that do not use each other's outputs, so their relative order is not fixed
C	Hiding unnecessary detail so that only the important features of a problem remain
D	A named, self-contained block of code that performs one task and can be called whenever it is needed
E	Where one step needs the output of an earlier step, so the order of the two steps is fixed
F	Running two or more tasks at the same time on a multi-core processor

2. A washing-machine control program runs each of the six steps below exactly once, but they are listed out of order. Number the steps 1–6 so that every dependency is respected (for example, water cannot be heated before it is in the drum, and the drum cannot spin until the water has gone). [3] (AO2)

Step	Order (1–6)
Spin	
Lock door	
Wash	
Fill with water	
Drain	
Heat water	

3. For each statement, write **True** or **False** in the middle column. If a statement is false, write a corrected version in the final column. [4] (AO2)

Statement	True / False	Correction if false
In a cinema-booking sequence <code>selectSeats</code> → <code>takePayment</code> → <code>issueTicket</code> , the step <code>issueTicket</code> could safely be moved before <code>takePayment</code> .		
Decomposing a large open-world game into modules lets each programmer in a team develop and test a different module at the same time.		
The washing machine also updates a countdown display; this update must always run after the wash step, because the order of every pair of tasks is fixed.		
If <code>gradeAnswer</code> is written once as a sub-procedure and called for every quiz question, a fix to the grading code only has to be made in one place.		

4. Complete the paragraph using words from the word bank. Three words are not needed. [2] (AO1)

When a problem is decomposed, each (a) _____ can be developed and tested separately. If one step uses the (b) _____ of an earlier step, there is a (c) _____ between the two steps and their order is (d) _____.

Word bank: sub-problem · output · dependency · fixed · input · abstraction · concurrent

5. A team is building an online cinema-ticket booking system. Identify **three** sub-procedures that the problem could be decomposed into. [3] (AO2)

Part B — Retrieval: keep it fresh (~6 marks)

6. (2.1.2) State what is meant by a **precondition**, and give one example of a precondition for a routine that performs a **binary search**. [2] (AO1/AO2)

7. (1.2.2) State, in order, the **two** stages of compilation that come **before** code generation. [2] (AO1)

8. (1.2.3) State **one** advantage of the **Agile** development methodology over the **Waterfall** lifecycle model. [2] (AO1)

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